

Master Thesis

Enabling Energy-Efficient Data Communication with Participatory Sensing and Mobile Cloud*

Sathyamoorthy PERAMANATHAN

January 9, 2014

Background

In the past few years, smart phones continued to become very popular and rapidly replacing most of the ordinary cellular phones. Among various reasons, new way of user interaction with touch screen, wider screen with high resolution, considerably many built-in applications support, typical comfortable support of web applications such as email, browser and revamped games are so attractive. But the development of batteries couldn't cope up with the rapid development of efficient mobile hardware components and that of wireless networks, limiting the mobile devices as resource-constraint in storage, intensive-computations, etc. In order to address these problems, *cloud computing* together with faster *wireless networks* were then considered and the integrated development gave rise to Mobile Cloud Computing(MCC). Nowadays, many popular mobile applications such as mobile social network, video streaming, email, instant messages are network-based and need to exchange data periodically with remote servers through cellular or WIFI. This network-based approach is optimistic with respect to resource-constraint mobile devices. But these applications are heavily used on mobile devices (e.g., mobile phones and tablets) and notorious for using up the battery energy quickly due to the large amount of data communications. Now this is a new kind of challenge in energy usage which needed to be optimized for better life of the mobile devices.

Keywords

Mobile Cloud Computing(mobile computing,cloud computing+wireless network), Proximate mobile computing entities, WIFI,3G, Energy-Efficient Communication, Crowd sourcing,Android

Introduction

In this project, we are going to address the energy-efficiency issue for mobile devices in data communication. Since wireless communication with poor signal strength consumes more energy, the idea is to hold non-time-sensitive traffic when the wireless communication signal strength is poor and then let the traffic go when the signal strength becomes strong. We will explore cloud-assisted approaches with participatory sensing to predict the signal quality of the mobile

*Partially developed content for mid course meeting Jan 2014

devices in different places and time. The data communication of mobile devices will then be optimized according to the signal quality from the environment. The goal of this project is to schedule data communication of mobile applications to achieve high energy-efficiency and prolong the lifetime of the mobile devices that is to let the data communication happen at right time with respect to the predictions about signal strength. We start with a cloud application to collect the necessary signal data continuously or periodically which will be used later for signal predictions. Using carefully designed energy-aware protocol we will have the control over data communications. Finally there is a strong intention to provide a tested and evolved version of efficient APIs package.

A simple, safer, harmless **cloud application** will be developed and provided to the volunteers who would like to participate anonymously in a crowd sourcing collection of signal information such as signal strength, time and location of their **mobile devices**. Among various cloud architectures, *centralized cloud* explained in [10] would be a suitable.

MultiNets [13] provides a simple yet powerful solution without additional network infrastructure support and new network protocols. The network switching architecture, MultiNets has *monitoring engine* as an independent component which is more suitable and extendable for this project's data collection part.

Then this huge collection of acquired data should be pre-processed and analyzed for two things: knowledge about *social behaviors* and corresponding influence in signal distribution, knowledge about *individual behaviors* to help the individual with **signal strength prediction** to access the data and media optimally with zero or smarter warnings.

As far as we know, all the solutions to energy efficiency in data communication so far provided, relies either on system level design such as real time network switching or optimal control in queuing networks using such as *Lyapunov optimization* [15]. Smart way of building knowledge is really important that may open up new ways of solving this problem. *Knowledge graphs* seems to be good option right now.

Using the knowledge body we will design an **energy-aware protocol**, the real agent in action, to optimize the data communication between cloud server and mobile devices.

Finally after tested on number of interesting and most occurring scenarios, it is most likely providing packaged APIs for other applications such as media downloader which requires energy-efficiency.

Literature

For the **background information** the literatures [3, 6, 12, 14, 16] can be referred. In order to get the idea about **crowd-sourcing** [1, 5] can be referred. Developing android **cloud applications for mobile devices** get started at Android Developer Site. For the different kind of existing works on **signal prediction and energy-aware protocols** [2, 4, 7–9, 11, 17] are good start. Finally the recent paper which suggest a strategy **eTime** for energy-efficiency in data communication between cloud and mobile devices will be analysed and compared against the proposed work of the thesis, however the thesis is not completely rely only on eTime [15]. After all it is one of the benchmarks and the project, specifically the strategies, will be driven by the insights and the knowledge learned from the acquired data.

Limitations

We have limitations in collecting the data of required quality and quantity in short span so in such case existing data sets may be used. As existing data is that it may lead heavy pre-processing, we still try to collect authenticate data with our application of whatever scale to get some insights. Though energy-efficient APIs will be provided for other application to use it. If time permits we will try to integrate on number of other applications.

References

- [1] John Carlo Bertot, Paul T Jaeger, and Justin M Grimes. Crowd-sourcing transparency: ICTs, social media, and government transparency initiatives. pages 51–58, 2010.
- [2] Hoang T. Dinh, Chonho Lee, Dusit Niyato, and Ping Wang. A survey of mobile cloud computing: architecture, applications, and approaches. *Wireless Communications and Mobile Computing*, 2011.
- [3] M. Frodigh, S. Parkvall, C. Roobol, P. Johansson, and P. Larsson. Future-generation wireless networks. *IEEE Personal Communications*, 2001.
- [4] Christine Jones, Krishna Sivalingam, Prathima Agrawal, and Jyh-cheng Chen. A survey of energy efficient network protocols for wireless networks. *Wireless Networks*, 2001.
- [5] S Kanhere. Participatory sensing: Crowdsourcing data from mobile smartphones in urban spaces. 2013.
- [6] Karthik Kumar, Jibang Liu, Yung-Hsiang Lu, and Bharat Bhargava. A survey of computation offloading for mobile systems. *Mobile Networks and Applications*, 18, 2012.
- [7] Young-Woo Kwon and E Tilevich. Energy-Efficient and Fault-Tolerant distributed mobile execution. pages 586–595, 2012.
- [8] Fangming Liu, Peng Shu, Hai Jin, Linjie Ding, and Jie Yu. Gearing resource-poor mobile devices with powerful clouds: Architecture, challenges and applications. *IEEE Wireless Communications*, jun 2013.
- [9] Kadathur S Natarajan and Chia-Chi Huang. Battery efficient operation of scheduled access protocol. 1993.
- [10] DI, NIU. GEARING RESOURCE-POOR MOBILE DEVICES WITH POWERFUL CLOUDS: ARCHITECTURES, CHALLENGES, AND APPLICATIONS. *IEEE Wireless Communications*, 2013.
- [11] Christopher Pluntke, Lars Eggert, and Niko Kiukkonen. Saving mobile device energy with multipath TCP. 2011.
- [12] Parthasarathy Ranganathan. Recipe for efficiency: principles of power-aware computing. *Communications of The ACM*, 2010.
- [13] NIRJON, SHAHRIAR, NICOARA, ANGELA, HSU, CHENG-HSIN, SINGH, JATINDER PAL, and STANKOVIC, JOHN A. *MultiNets: A System for Real-Time Switching between Multiple Network Interfaces on Mobile Devices*. 2012.

- [14] Ashish Sharma, Vishnu Navda, Ramachandran Ramjee, Venkata Padmanabhan, and Elizabeth Belding. Cool-Tether: energy efficient on-the-fly wifi hot-spots using mobile phones. 2009.
- [15] Peng Shu, Fangming Liu, H Jin, M Chen, Feng Wen, Yupeng Qu, and B Li. eTime: energy-efficient transmission between cloud and mobile devices. pages 14–19, 2013.
- [16] I. Stojmenovic. Handbook of wireless networks and mobile computing. 2002.
- [17] Sherali Zeadally, Samee Khan, and Naveen Chilamkurti. Energy-efficient networking: past, present, and future. *The Journal of Supercomputing*, 0.